# SciBite: A bespoke approach to facilitating FAIR practice in the Life Sciences Industry

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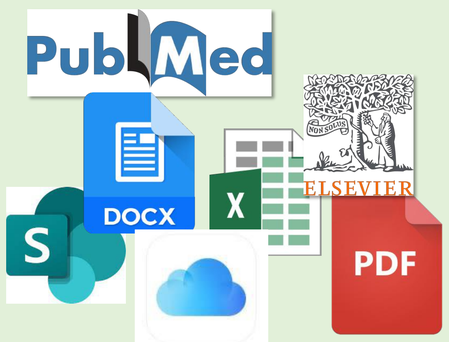
## Problem

**Companies have multiple information-rich documents in various formats and locations making it difficult to find relevant information using simple search techniques**



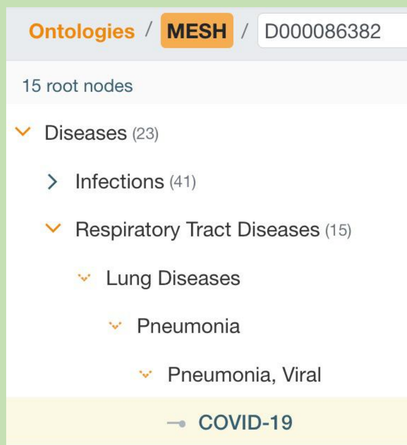

- Want to find all documents relevant to a search, e.g. *Covid-19 treatments*
- Want to find answers to questions, e.g. *What drugs are used to treat Covid-19?*
- Unstandardised data entry makes searching tedious, e.g. *CT-P59, Regdanvimab, Regkirona are synonyms of the same drug*

NOT Findable Accessible Interoperable Reusable

## SciBite Solutions

**Making data** Findable Accessible Interoperable Reusable

### CENtree

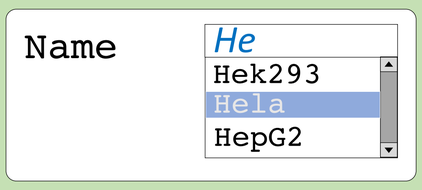


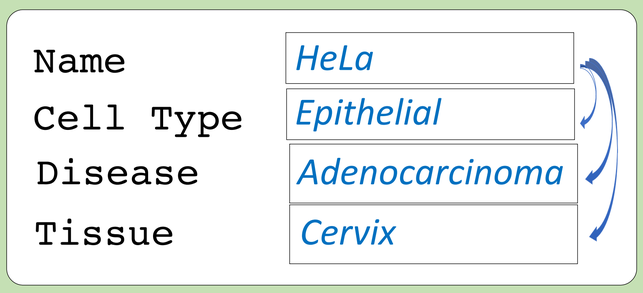

### Ontology management

- Expand public ontologies
- Create custom ontologies
- Ontologies can be used to create VOCabs

**Use case:** Standardised data entry in Electronic Lab Notebooks and Registration Systems

**Smart Forms**




Type ahead > Select term > Class properties are auto-filled

### VOCabs

**Expertly curated vocabularies**

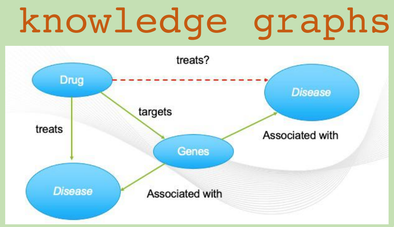
- Manually curated
- Domain specific, e.g. drug, disease
- Use public and/or custom entities
- Add missing classes
- Add synonyms incl. automated synonym expansion
- Add context to synonyms
- Reduce ambiguity
- Optimise vocabularies for named entity recognition

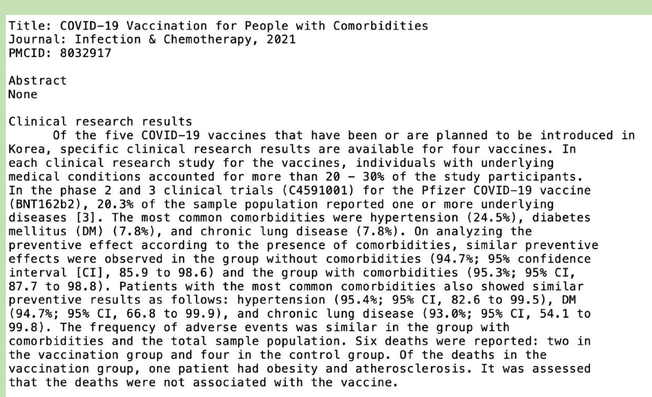
### TERMite

**Text Analysis Engine**

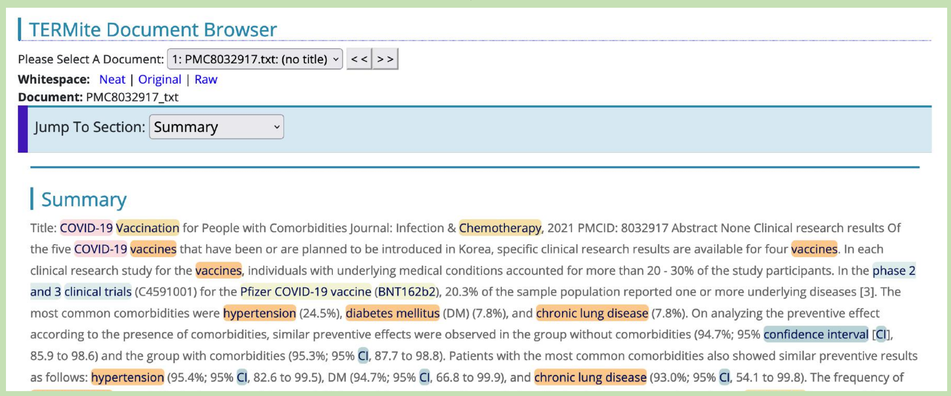
- Uses curated VOCabs to transform scientific text into machine-readable data
- Maps unstructured text to standardised entities

**Use case:** Standardised machine-readable knowledge graphs




**Use case:** Find all documents containing drugs and diseases




**Text is mapped to public ontology classes where possible. If no public class exists, a custom one is made.**

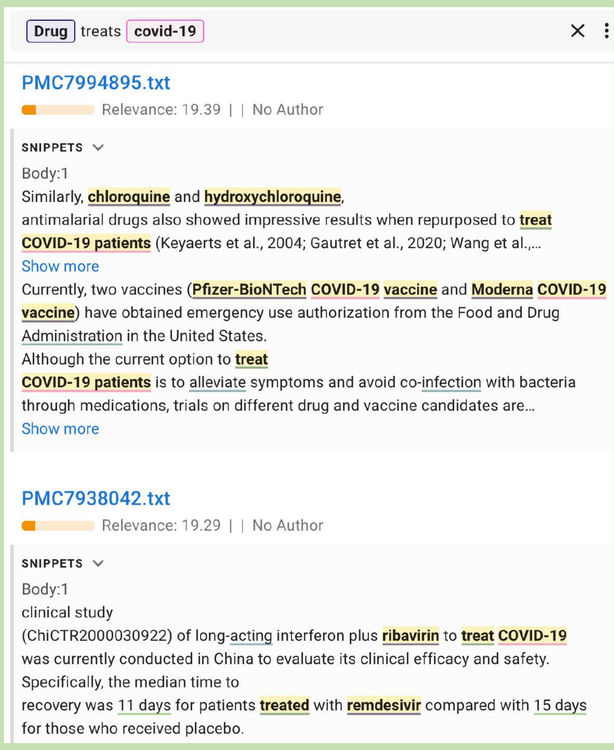
SciBite custom | ChEMBL | MeSH

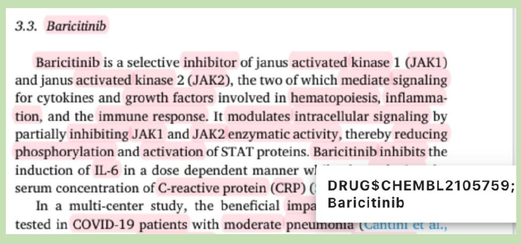
| | | | |
|---|---|---|---|
| DRUG | COVID-19 Vaccine Janssen | SDG02022 | Ad26.COV2.S |
| DRUG | Moderna COVID-19 vaccine | CHEMBL4650491 | Moderna COVID-19 vaccine; mRNA-1273 |
| INDICATION | HIV Infections | D015658 | infected with human immunodeficiency; HIV; human immunodeficiency viru s; HIV-infected; HIV infection |
| INDICATION | Lung Diseases | D008171 | lung disease; lung disease |
| INDICATION | Atherosclerosis | D050197 | atherosclerosis |
| INDICATION | COVID-19 | D000086382 | COVID-19 infection; COVID-19; infected with COVID-19 |

### SciBite Search

**Search and Analytics**

- Can load proprietary or public documents
- Documents are TERMite-processed to enrich unstructured text
- Complex searches can answer questions
- PDF markup




**Use case:** What drugs are used to treat Covid-19?

### SciBite AI

**Machine Learning Models**

- NER models
- Relationship extraction
- Question/Answering

SciBite products are sold under license, but we contribute to community projects, such as;

- Pistoia Alliance
- Covid-19 data challenge
- Feedback to public ontologies

Email: help@scibite.com